

Project Specification: Specialized Speech-to-Text for Radiology

1. Executive Summary

This document outlines the technical requirements and architecture for a specialized Speech-to-Text (STT) service designed specifically for radiologists. The system aims to provide high-accuracy, real-time transcription of medical diagnostic findings, leveraging a locally hosted instance of OpenAI's Whisper model fine-tuned for medical and radiological terminology.

2. Core Functional Requirements

2.1 Live Audio Capture

- Real-time Stream:** The interface must support low-latency audio capture from professional-grade microphones.
- Local Processing:** To maintain patient privacy and data security, all audio processing and transcription must occur locally on-site.

2.2 Specialized Transcription Engine

- Local Whisper Deployment:** Integration of the Whisper model (e.g., "medium" or "large-v3" variants) running on local GPU resources.
- Radiology Lexicon:** The model must be fine-tuned or augmented with specialized datasets containing radiological terminology (e.g., anatomical terms, pathology descriptors, and modality-specific jargon like "hyperechoic," "T1-weighted," or "pneumothorax").
- Contextual Accuracy:** Handling of medical abbreviations and acronyms commonly used in clinical reports.

3. Technical Architecture

Component	Technology/Method
STT Model	OpenAI Whisper (Local Instance)
Fine-Tuning/Optimization	LoRA (Low-Rank Adaptation) or medical-specific prompt engineering
Hardware Target	NVIDIA GPU with CUDA support for accelerated inference

Component	Technology/Method
Audio Processing	FFmpeg for real-time stream normalization and chunking

4. Implementation Workflow

1. **Input:** Radiologist activates the recording and speaks findings.
2. **Buffering:** Audio is streamed in short chunks to the local Whisper inference engine.
3. **Transcription:** The engine applies the specialized radiology weights to convert audio to text.
4. **Review:** The transcribed text is displayed immediately for real-time verification and editing.

5. Security and Compliance

By utilizing a local deployment of the model, the service inherently minimizes the risk of data breaches associated with cloud-based processing. The system will be designed to align with HIPAA and other healthcare data protection standards by ensuring no patient-identifiable audio data leaves the local network.